Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.

The retriever finds relevant documents from a knowledge base. The language model then uses the retrieved information to formulate an answer.

RAG reduces hallucinations because answers are grounded in real source documents.

How Chunking and Embeddings Work

Long documents are split into smaller chunks, typically around 500 tokens with some overlap between neighbouring chunks.

Each chunk is converted into a numerical vector by an embedding model. Chunks with similar meaning have vectors that are close together.

The vectors are stored in a FAISS index, which supports fast similarity search.